STUDENT PERFORMANCE ANALYSIS AND PREDICTION SYSTEM

Minor Project

Training Program: AI/ML with Python Training Program

Training Institute: InternsElite

Batch: September 2026

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Submitted as part of the AI/ML with Python Training Program.

Project Title and Student Details

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Project Overview

The Student Performance Analysis and Prediction System is a machine learning project developed using Python. It analyses students' academic and personal information and uses a Linear Regression model to estimate their final mathematics examination grades.

The project includes data preprocessing, exploratory data analysis, data visualization, model training, prediction, and performance evaluation.

Introduction

Student academic performance is influenced by several factors, including previous examination results, study habits, attendance, family background, and other personal and academic characteristics.

Understanding these factors can help identify patterns in student performance. Data analysis and machine learning provide useful techniques for examining historical academic data and estimating future outcomes.

This project uses the Student Performance dataset from the UCI Machine Learning Repository. Python libraries such as Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn are used to analyse the dataset and develop a prediction model.

The system uses Linear Regression to estimate the final mathematics grade, represented by the variable G3.

Problem Statement

Educational performance data contains several attributes that may be related to students' examination results. Analysing these attributes manually can be difficult, especially when working with multiple variables.

The main problem addressed by this project is how to analyse student academic data, identify meaningful patterns, and use machine learning to estimate final examination grades.

The system aims to provide a data-driven understanding of academic performance. Its predictions are estimates and should not be treated as guaranteed examination results.

Project Objectives

The main objectives of this project are:

1. To load and understand a real-world student performance dataset.
2. To inspect dataset structure, columns, data types, and missing values.
3. To clean and prepare the dataset for analysis.
4. To calculate descriptive statistics.
5. To perform exploratory data analysis.
6. To create visualizations representing academic performance patterns.
7. To identify relationships between student-related features and final grades.
8. To develop a Linear Regression model for predicting final mathematics grades.
9. To evaluate the model using MAE, RMSE, and R^2 score.
10. To apply cross-validation and basic hyperparameter tuning.
11. To compare actual and predicted examination marks.
12. To interpret the results and identify possible improvements.

Dataset Description

1. Dataset Description

This project uses the Student Performance Mathematics dataset, named student-mat.csv.

The dataset contains information about students, including demographic details, family background, study habits, school-related information, and examination grades.

The dataset contains:

- **Total records:** 395
- **Total columns:** 33

- **Target variable:** G3
- **Target meaning:** Final mathematics examination grade
- **Grade range:** 0 to 20

The study-time variable is recorded in categories rather than exact hours.

Attendance-Related Information

The dataset contains an absences variable that records the number of school absences for each student. It does not provide an attendance percentage. Therefore, absences are used as the available attendance-related feature in this project.

2. Important Dataset Attributes

Some important attributes used in the project include:

Attribute	Description
Age	Student's age
sex	Student's sex
studytime	Weekly study-time category
failures	Number of past class failures
absences	Number of school absences
G1	First-period grade
G2	Second-period grade
G3	Final mathematics grade

The model uses the available input features except G3, which is the target variable.

Dataset Source

Source: UCI Machine Learning Repository

Dataset: Student Performance

Website: https://archive.ics.uci.edu/dataset/320/student+performance

The mathematics dataset file used in this project is student-mat.csv.

Data Preprocessing and Feature Engineering

Data preprocessing is an important step in machine learning. It helps prepare raw data for analysis and model training.

The following preprocessing steps were implemented:

1. Loading the Dataset

The dataset was loaded using the Pandas library. Since the original CSV file uses semicolons as separators, the file was loaded with the appropriate separator setting.

2. Dataset Inspection

The dataset was inspected to understand:

- Number of rows and columns.
- Column names and data types.
- Missing values.
- Duplicate records.
- Target variable values.

3. Data Cleaning

The project checked for duplicate records and removed duplicates where applicable. The target variable G3 was converted to numeric values, and records with an invalid or missing target were excluded.

The final dataset contained 395 records. No duplicate records were removed, and no missing values were found in the dataset used for this run.

4. Handling Numerical and Categorical Features

The input features were divided into numerical and categorical groups.

A preprocessing pipeline was created using Scikit-learn:

- Numerical features: median imputation.
- Categorical features: most-frequent imputation followed by one-hot encoding.

Although the current dataset did not contain missing values, the imputation steps make the pipeline more suitable for handling missing values in future datasets.

Two additional features were created during feature engineering:

- `previous_grade_average`: Calculated as the average of G1 and G2 grades.
- `absence_count`: Represents the number of school absences using the existing absences column.

These features help provide additional information for analysing student academic performance.

5. Feature and Target Selection

The target variable was selected as G3, representing the final mathematics grade.

All other available columns were used as input features.

The target variable was kept separate from the input features to prevent it from being used to predict itself.

Exploratory Data Analysis and Visualizations

Exploratory Data Analysis (EDA) was performed to understand the dataset and identify patterns in student academic performance.

Descriptive statistics were calculated, and visualizations were generated using Matplotlib and Seaborn.

1. Descriptive Statistics

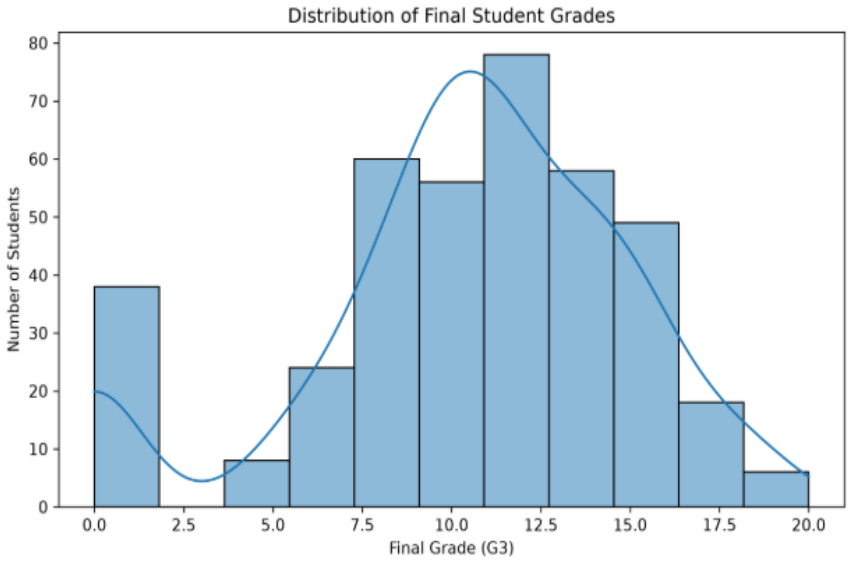
Important statistical observations from the dataset are:

Variable	Mean	Median	Minimum	Maximum
G3	10.42	11	0	20
G1	10.91	11	3	19
G2	10.71	11	0	19
Study Time Category	2.04	2	1	4
Absences	5.71	4	0	754

The average final mathematics grade (G3) is approximately 10.42 out of 20, while the median is 11.

2. Final Grade Distribution

A histogram was created to understand how final mathematics grades are distributed among students.

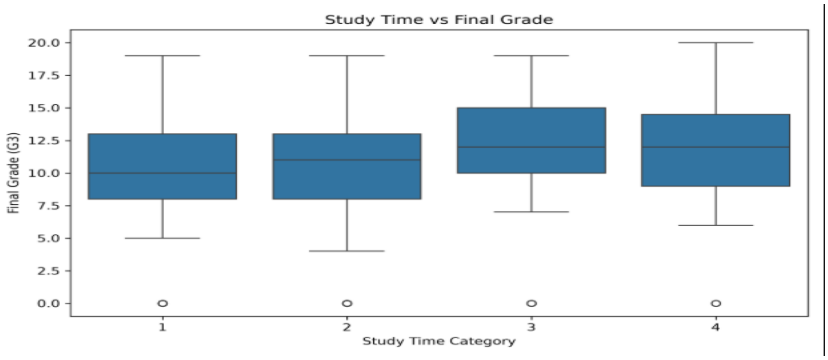


Final Grade Distribution

Interpretation: The histogram shows the distribution of students' final mathematics grades. It helps identify the concentration of grades and the variation in student performance.

3. Study Time and Final Grade

A box plot was created to compare final grades across different study-time categories.

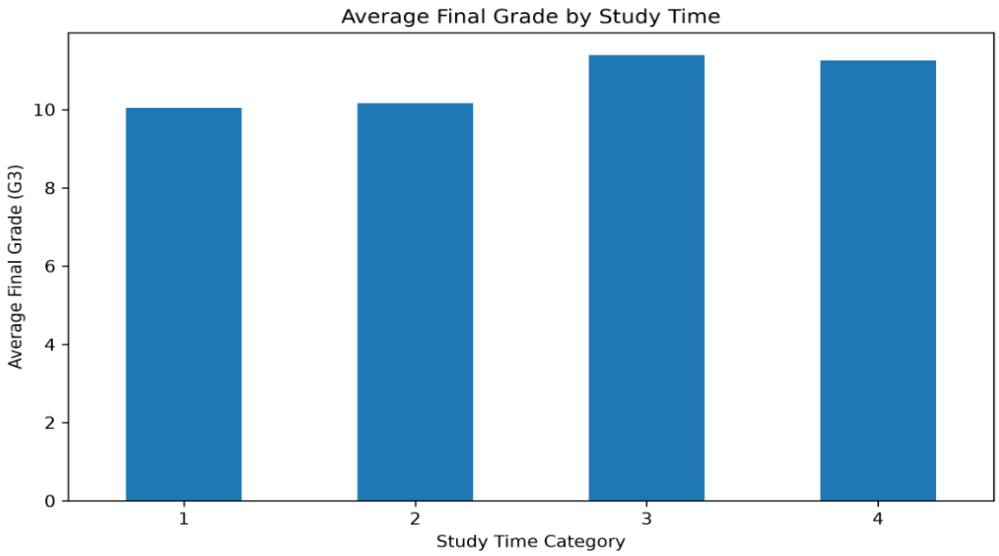


Study Time and Final Grade

Interpretation: This visualization helps compare the distribution of final grades among students belonging to different study-time categories. It also shows the variation in grades within each category.

4. Average Grade by Study Time

A bar chart was generated to compare the average final grade across study-time categories.

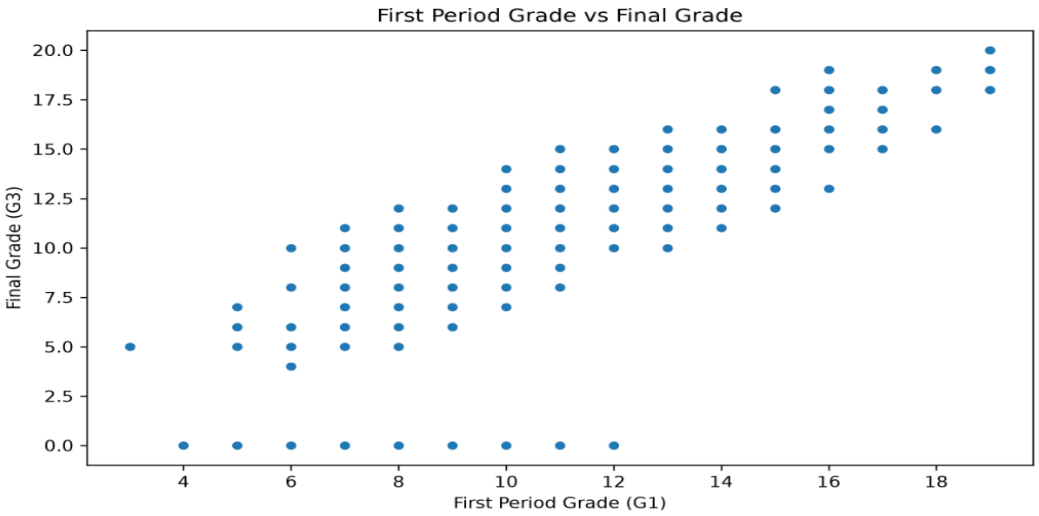


Average Grade by Study Time

Interpretation: The bar chart provides a comparison of average academic performance across the different study-time categories.

5. First-Period Grade vs Final Grade

A scatter plot was generated to examine the relationship between first-period grades (G1) and final grades (G3).

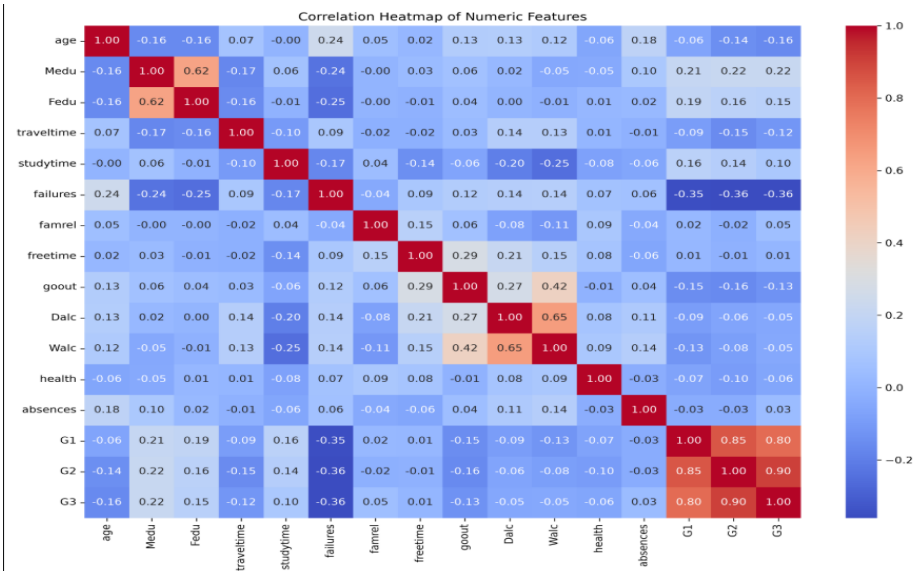


First-Period Grade vs Final Grade

Interpretation: The visualization shows a positive relationship between G1 and G3. Students with higher first-period grades generally tend to have higher final grades in this dataset.

6. Correlation Heatmap

A correlation heatmap was generated to examine relationships between numerical variables.



Correlation Heatmap

Interpretation: The heatmap helps identify positive and negative relationships between numerical features. It provides a visual overview of how different numerical attributes are associated with one another.

Machine Learning Methodology

1. Machine Learning Algorithm

Linear Regression was selected as the machine learning algorithm for predicting final mathematics grades.

Linear Regression is a supervised learning algorithm used to predict a numerical value based on relationships between input features and a target variable.

In this project:

- Input: Student-related and academic features.
- Output: Predicted final mathematics grade (G3).
- Algorithm: Linear Regression.

2. Preprocessing Pipeline

A Scikit-learn preprocessing pipeline was implemented to handle numerical and categorical data.

The numerical and categorical preprocessing steps were combined using a ColumnTransformer.

The preprocessing pipeline was integrated with the regression model so that preprocessing was fitted using training data within the model-training workflow.

This approach helps reduce the risk of data leakage.

3. Model Selection

Linear Regression was selected because the project focuses on predicting a continuous numerical value.

It also provides a simple baseline model that can be evaluated using standard regression metrics.

Model Training and Testing Approach

1. Train-Test Split

The dataset was divided into training and testing subsets.

The project used an 80:20 train-test split.

Dataset	Number of Records
Training Data	316
Testing Data	79
Total	395

A random state of 42 was used to make the split reproducible.

2. Model Training

The Linear Regression model was trained using the training dataset.

The preprocessing pipeline was fitted as part of the training process. The trained model was then used to generate predictions for the testing dataset.

3. Cross-Validation

Five-fold cross-validation was used to examine model performance across different subsets of the training data.

This helps provide a more consistent estimate of model performance than relying only on a single train-test split.

4. Hyperparameter Tuning

Basic hyperparameter tuning was performed using GridSearchCV.

The fit_intercept parameter was tested with two possible values:

- True
- False

The best parameter selected by the tuning process was:

Best Parameter: fit_intercept = True

The best tuning cross-validation RMSE was approximately 2.0195.

Evaluation Metrics and Results

The model was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 score.

Cross-validation results were also recorded.

1. Evaluation Metrics

Mean Absolute Error (MAE): Measures the average absolute difference between actual and predicted values. Lower values indicate smaller prediction errors.

Root Mean Squared Error (RMSE): Measures prediction error while giving greater weight to larger errors. Lower values indicate better performance.

R^2 Score: Measures how much of the variation in the target variable is explained by the model. A higher value generally indicates a better fit.

2. Model Evaluation Results

The following results were obtained during the final project execution:

Metric	**Result**
Total Records	395
Training Records	316
Testing Records	79
MAE	1.6467
RMSE	2.3784
R^2 Score	0.7241
Cross-Validation Mean R^2	0.8039
Cross-Validation Standard Deviation	0.0317
Best Tuning CV RMSE	2.0195
Best fit_intercept	True

3. Result Interpretation

The model achieved an MAE of approximately 1.65 marks and an RMSE of approximately 2.38 marks on the test dataset.

The R^2 score of 0.7241 indicates that the model explains approximately 72.41% of the variation in final grades on the test dataset.

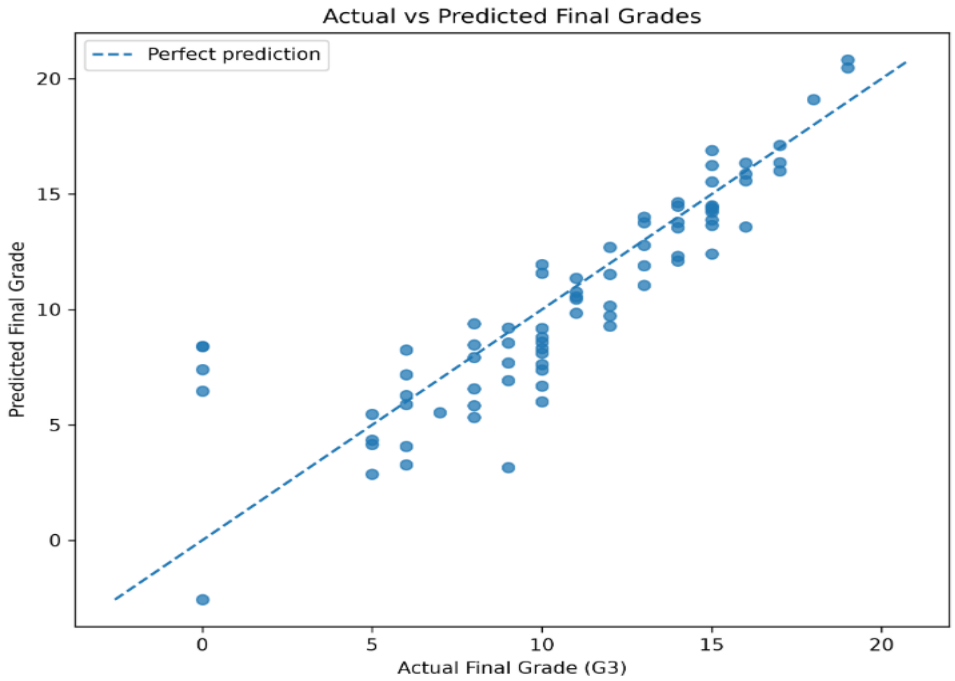
The five-fold cross-validation mean R^2 score was approximately 0.8039, with a standard deviation of 0.0317.

These results indicate that the model learned useful patterns from the available student data. However, its predictions are estimates and may differ from actual examination results.

Actual Versus Predicted Examination Marks

After training, the model generated predicted final grades for the test dataset.

The actual and predicted grades were compared using a visualization.



Actual vs Predicted Grades

Interpretation:

The actual-versus-predicted visualization helps compare the grades achieved by students with the grades estimated by the Linear Regression model.

Points closer to the ideal agreement line represent predictions that are closer to the actual grades. Differences between actual and predicted values represent prediction errors.

This comparison helps evaluate how well the model performs on unseen testing data.

Interpretation of Findings and Academic Performance Patterns

The analysis produced several useful observations:

1. The average final mathematics grade was approximately 10.42 out of 20.
2. The median final mathematics grade was 11.
3. The dataset showed variation in final grades among students.
4. First-period grades (G1) showed a positive relationship with final grades (G3).
5. Study-time categories showed differences in the distribution and average of final grades.
6. The correlation heatmap helped identify relationships between numerical academic and student-related features.
7. The Linear Regression model achieved an R^2 score of 0.7241 on the test dataset.
8. The correlation between second-period grades (G2) and final grades (G3) was 0.905, indicating a strong positive relationship.
9. The correlation between absences and final grades (G3) was 0.034, indicating almost no linear relationship.

10. The average final grades (G3) for study-time categories 1, 2, 3, and 4 were 10.05, 10.17, 11.40, and 11.26 respectively.

These observations describe patterns within the dataset. They do not prove that any single factor directly causes better or worse academic performance.

Limitations

Although the project successfully performs student performance analysis and prediction, it has some limitations:

1. The dataset contains only 395 student records.
2. The dataset represents a particular educational context and may not represent every school or student population.
3. Linear Regression may not capture complex non-linear relationships.
4. Predicted grades are estimates and are not guaranteed to match actual examination results.
5. Study time is recorded in categories rather than exact study hours.
6. The model uses G1 and G2 as input features. Therefore, it is more suitable for estimating final grades when earlier examination results are already available.
7. The model has not been tested on a separate external dataset.
8. The dataset provides the number of school absences but does not contain an attendance-percentage field. Therefore, attendance percentage analysis could not be performed.

Future Improvements

The project can be improved in several ways:

1. Use larger and more diverse student datasets.
2. Compare Linear Regression with other regression algorithms.
3. Explore additional feature engineering techniques.
4. Perform more extensive hyperparameter tuning.
5. Evaluate the model using additional test datasets.
6. Develop a user-friendly web application for entering student information and viewing predictions.
7. Explore a separate pass/fail classification model as an optional extension.

Conclusion

The Student Performance Analysis and Prediction System successfully demonstrates the use of Python, data analysis, visualization, and machine learning for analysing student academic performance.

The project used the UCI Student Performance Mathematics dataset containing 395 records and 33 columns. Data preprocessing and exploratory data analysis were performed to understand the dataset and identify academic performance patterns.

A Linear Regression model was developed to estimate final mathematics grades. The model was trained and tested using an 80:20 split, and five-fold cross-validation and basic hyperparameter tuning were also applied.

The final model achieved an MAE of 1.6467, an RMSE of 2.3784, and an R^2 score of 0.7241 on the test dataset.

Overall, the project provided practical experience in data cleaning, exploratory data analysis, data visualization, machine learning model development, prediction, and evaluation.

The project also highlights the importance of using data-driven approaches to understand academic performance. The knowledge gained from this work can be useful for developing more advanced student performance analysis and prediction systems in the future.

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